ELECTRONIC VOLTMETERS

1. D.C VOLTMETERS

D.C voltmeters consists of d.c amplifier and an attenuator. D.C voltmeter is divided into two categories.

(i) Direct-coupled amplifier D.C voltmeter

(ii) Chopper type D.C voltmeter

Direct-coupled amplifier D.C voltmeter

This type of voltmeter is used to measure voltages of the order of millivolts owing to limited amplifier gain.

The circuit diagram of a direct-coupled amplifier d.c voltmeter using cascaded transistors is shown in figure below.

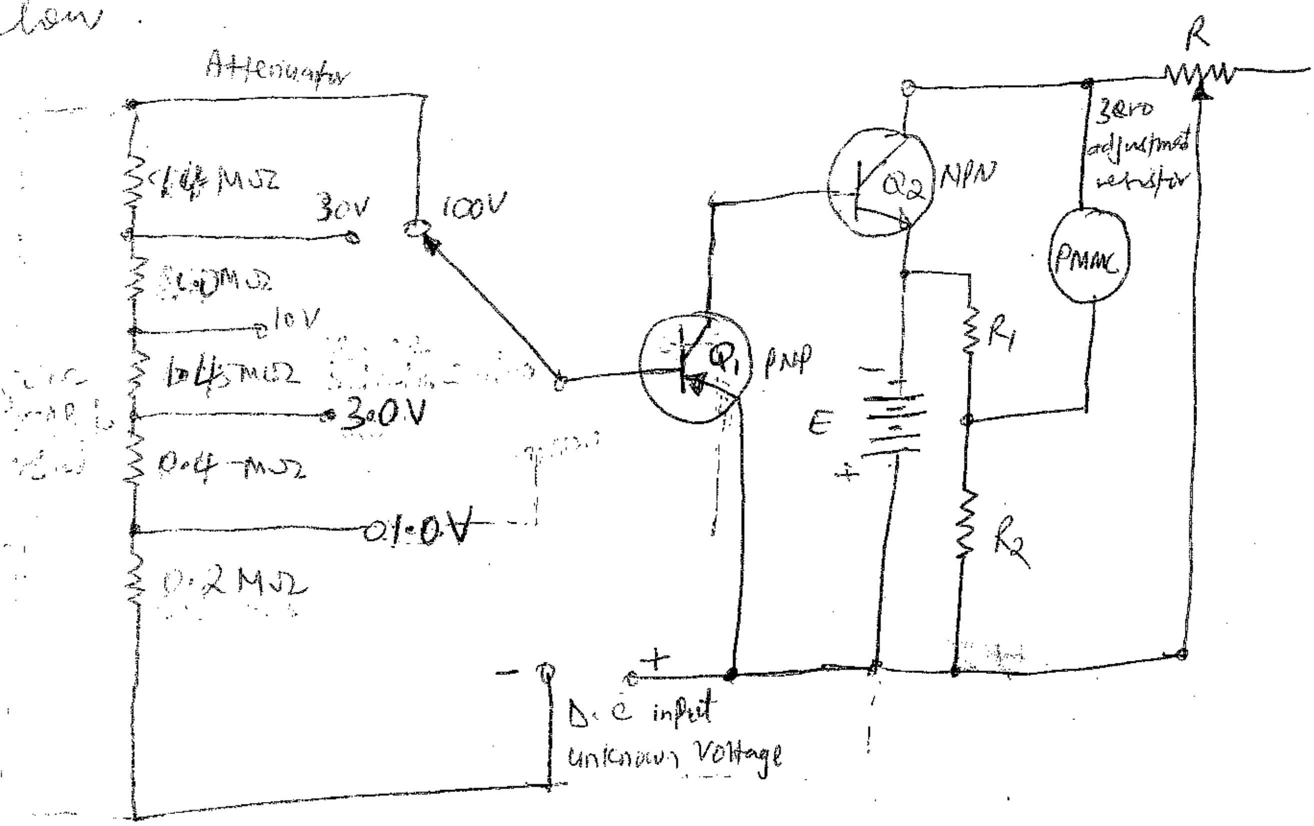


Figure: Direct Coupled Amplifier D.C voltmeter using Cascaded Transistors

An attenuator is used in input stage to select voltage range. A transistor is a current controlled device so resistance is inserted in series with the transistor Q_1 to select the voltage range.

Two transistors in cascaded connections are used instead of using a single transistor for amplification in order to keep the sensitivity of the circuit high. Transistors Q_1 and Q_2 are taken complement to each other and are directly coupled to minimize the number of components in the circuit. They formed a direct coupled amplifier.

A variable resistor R is put in the circuit for zero adjustment of the PMMC. In this voltmeter voltage to be measured is firstly attenuated with the range selector switch to keep the input voltage of amplifier within its level. The voltage under measurement is amplified by the direct coupled dc amplifier whose output is further supplied to PMMC meter. When input voltage exceeds its limit, amplifier get saturated which limits the current passing through the PMMC meter so meter does not burn out.

CHOPPER TYPE D.C VOLTMETER

Chopper type d.c amplifier is used in highly sensitive d.c electronic voltmeters. Its block diagram is as shown in figure

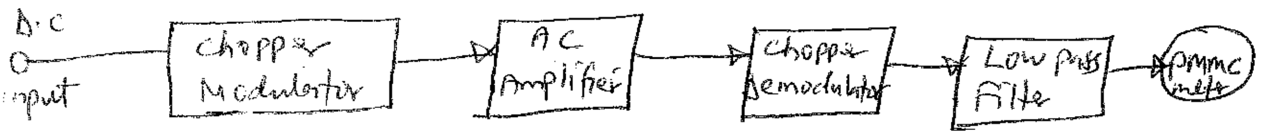


Figure Block diagram of chopper type D.C voltmeter

Firstly, d.c input voltage is converted into a.c voltage by chopper modulator and then it is supplied to an a.c amplifier. Output of amplifier is then demodulated to a d.c voltage proportional to the original input voltage. Modulator chopper and demodulator act in antisynchronism.

A.C VOLTMETERS

Electronic a.c voltmeters are basically identical to d.c voltmeters except that the a.c input voltage must be rectified before it can be applied to the d.c meter circuit. In some instances, rectification takes place before amplification, in which case a simple diode rectifier circuit precedes a d.c amplifier as shown in figure below.

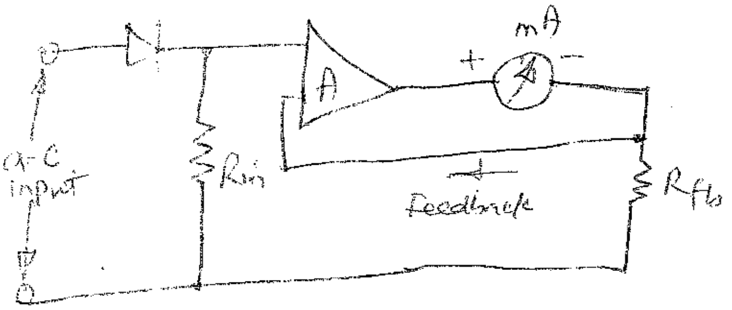


Figure (a) d.c mode of operation

In another approach the a.c signal is rectified after amplification as shown in figure below, where full-wave rectification takes place in the meter circuit connected to the output terminals of the a.c amplifier.

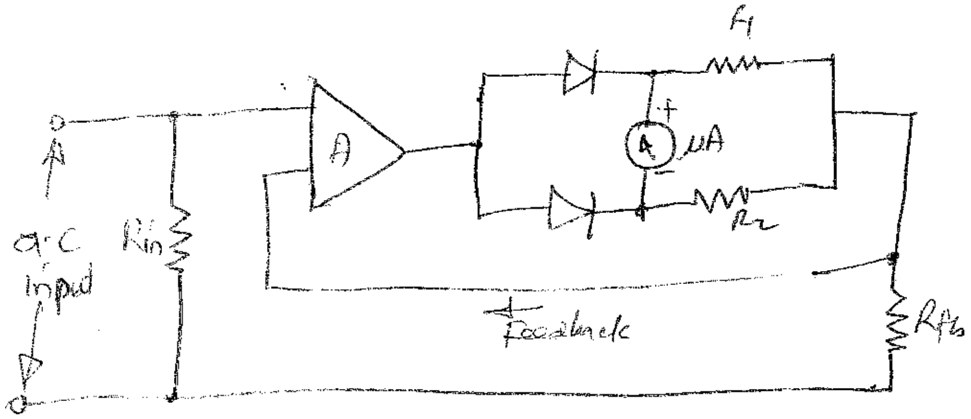


Figure (b) a.c mode of operation

This approach requires an a.c amplifier with a high open loop gain and large amount of negative feedback to overcome the non-linearity of the rectifier diodes.

A.C VOLTMETERS USING RECTIFIERS

1. HALF WAVE

The series connected diode provide half-wave rectification and the average value of the half-wave voltage is developed across the resistor and applied to the input terminals of the d.c amplifier as shown shown in figure

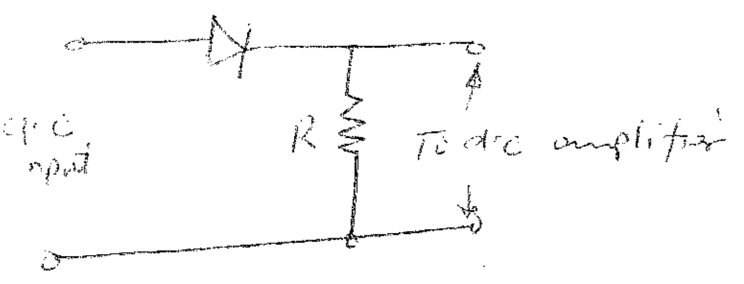


Figure (9) Series-connected diode.

2. FULL-WAVE RECTIFICATION

Full-wave rectification can be obtained by the bridge circuit shown in figure, where the average value of the sine wave is applied to the amplifier and meter circuit.

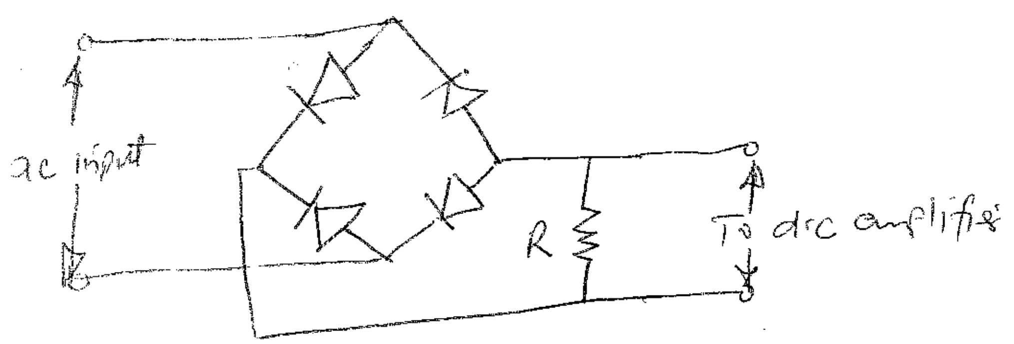


Figure Full-wave rectification

3. SHUNT-CONNECTED DIODE

In some cases, there may be requirement to measure the peak value of a waveform instead of the average value, a circuit as shown in figure is used.

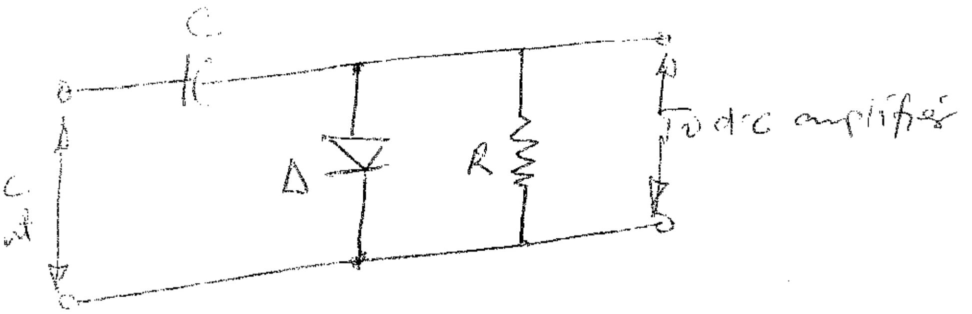


Figure Shunt-Connected diode

In this circuit the rectifier diode changes the small capacitor to the peak of the applied input voltage and the meter will ~~therefore~~ indicate the peak voltage. In most cases, the meter scale is calibrated in terms of both the rms and peak values of the sinusoidal input waveform.
Nonsinusoidal waveforms when applied to this voltmeter, will cause the meter to read either high or low, depending on the form factor of the waveform.

DIFFERENTIAL VOLTMETERS

One of the most accurate methods of measuring an unknown voltage is the differential voltmeter ~~tee~~ technique, where the voltmeter is used to indicate the difference between the unknown voltage and a known voltage. This type of instrument is sometimes called a potentiometric voltmeter. The differential voltmeter is shown in figure

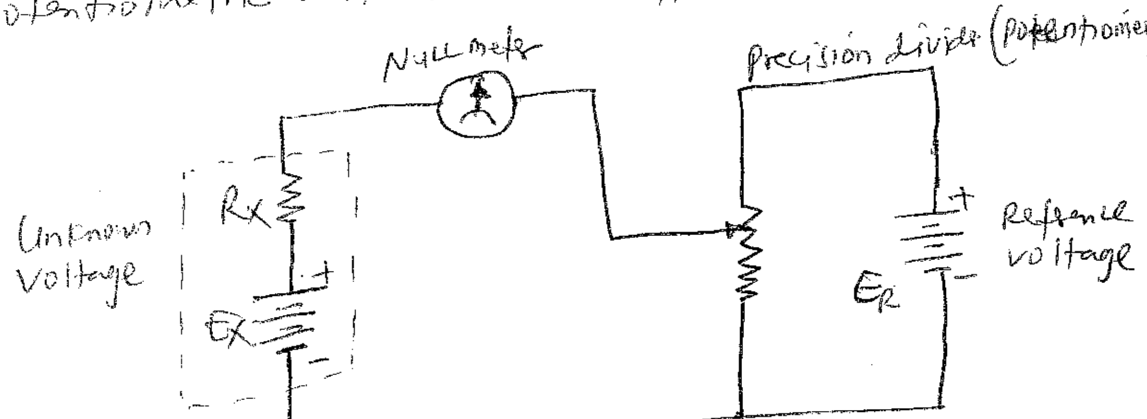


Figure Basic differential voltmeter

In this circuit, the null meter is connected between the unknown voltage source and the output terminals of a precision voltage divider, and hence indicates the difference between the two. This voltage divider is connected across a reference voltage source and can be adjusted to provide accurately known fractions of the reference voltage.

In performing a measurement, the divider is adjusted until its output voltage equals the unknown voltage. The null meter, which is connected between the unknown source and the divider output terminals, indicates zero volts when the two voltages are equal. In this null condition neither the source nor the reference supplies current to the meter, and the differential voltmeter therefore presents an infinite impedance to the source under test.

DIGITAL INSTRUMENTS

Digital instruments use logic circuits and techniques for carrying out measurement of quantities. Digital instruments have the advantage of easy readability of the measurement results because of the digital readout. The digital instruments provides ~~accurate~~ accurate reading than the analog type. Digital instruments provide better resolution than analog instruments. They have greater speed than analog ones. Digital instruments have digital readout. Virtually all digital instruments provide a digital display of the measurand.

Digital instruments such as digital voltmeters or multimeters are used for measurement of analog quantities. Conversion of an analog signal into an equivalent digital signal is, therefore, essential. Thus analog-to-digital converters (ADCs) are most important parts of digital instruments.

1. DIGITAL VOLTMETERS (DVMs)

The digital voltmeter (DVM) displays measurements of ac or dc voltages as discrete numerals instead of a pointer deflection on a continuous scale as in analog instruments. Numerical readout is advantageous in many applications because it reduces human reading and interpolation errors, eliminates parallax error, increases reading speed, and often provides output in digital form suitable for further processing and recording.

The DVM is a versatile and accurate instrument that can be used in many laboratory measurement applications. Since the the development and perfection of integrated circuits (IC) modules, the size, power requirements, and cost of the DVM have been drastically reduced so that DVMs can compete with the

conventional analog instruments, both in portability and price.
Digital voltmeters can be classified according to the following broad categories:

(i) Ramp-type DVM
(ii) Integrated DVM.
(iii) Continuous-balance DVM
(iv) Successive-approximation DVM

1. RAMP TYPE DVM

The operating principle of the ramp-type DVM is based on the measurement of the time it takes for a linear ramp voltage to rise from 0V to the level of the input voltage, or to decrease from the level of the input voltage to zero.

The operating principle and block diagram of a ramp-type DVM are given in Figure.

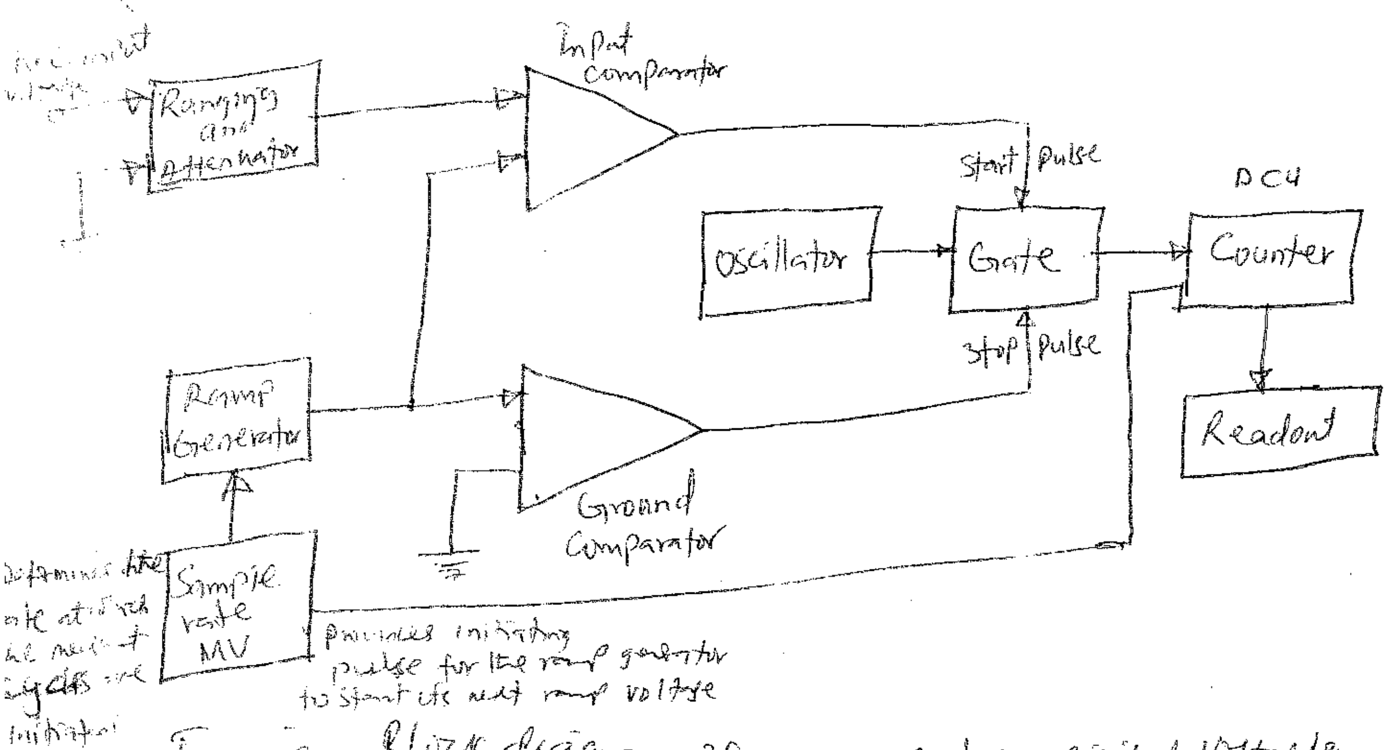


Figure Block diagram of a ramp-type digital voltmeter

At the start of the measurement an oscillator generates clock pulses which are allowed to pass through the gate to a number of decade counting units (DCUs) which totalize the number of pulses passed through the gate. The decimal number displayed by the indicator tubes associated with the DCUs, is a measure of the magnitude of the input voltage.

The sample-rate multivibrator determines the rate at which the measurement cycles are initiated. The oscillation of this multivibrator can usually be adjusted by a front panel control, marked rate, from a few cycles per second to as high as 1,000 or more. The sample-rate circuit provides an initiating pulse for the ramp generator to start its next ramp voltage. At the same time, a reset pulse is generated which returns all the DCUs to their 0 state, removing the display momentarily from the indicator tubes.

2. INTEGRATING TYPE DVM

Such a digital voltmeter makes use of an integrating technique which employs a voltage to frequency (V/F) conversion. This voltmeter indicates the true average value of the unknown voltage over a fixed measuring period.

3. CONTINUOUS-BALANCE DVM

The continuous-balance DVM instruments provides excellent performance. The block diagram of a servo-driven continuous balance DVM is given in the figure. The dc input voltage ~~is voltage~~ is applied to an input attenuator that provides suitable range switching. The input attenuator is a front panel control that also carries a decimal point indicator to move on the display area in accordance with the input range selected. After passing through an over voltage protection circuit and a-c rejection filter, the input voltage is applied to one side of a mechanical chopper comparator. The other side of the comparator is connected to the wiper arm of the motor driven precision potentiometer, connected across a reference supply. The output of the chopper-comparator, which is driven by the line voltage and vibrate at the line frequency rate, is a square wave signal. The amplitude of the square wave is a function of the difference in magnitude and polarity of the dc voltages connected ~~connected~~ to the opposite sides of the chopper. The square wave of the signal is applied ~~by~~ to a high impedance, low noise preamplifier and fed to a power amplifier. The servo motor on receiving the amplified square wave difference signal, drives the arm of the precision potentiometer in the direction required to cancel the difference voltage across the chopper comparator. The servo motor also drives a drum-type mechanical indicator that has the digits 0 to 9 imprinted about the periphery of its drum segments. The position of the servo motor shaft corresponds to the amount of feedback voltage required to null the chopper input, and this position is indicated by the drum-type indicator. The position of the shaft therefore is an indication of the magnitude of the input voltage.

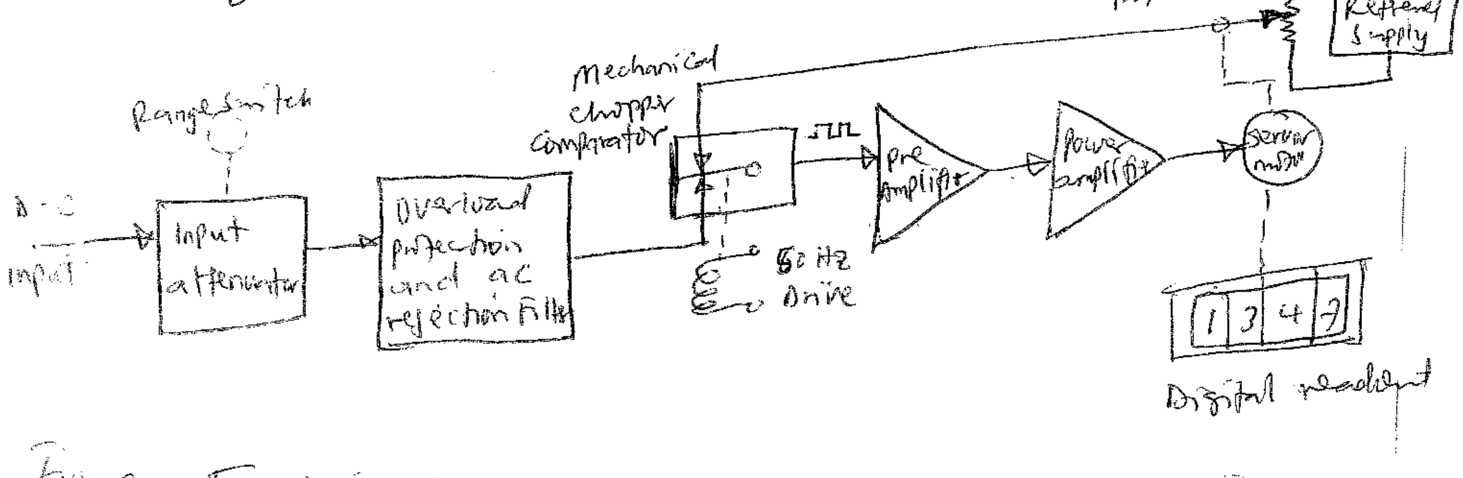


Figure Functional Block diagram of a servo-balancing potentiometric type digital voltmeter

SUCCESSIVE-APPROXIMATION DVM

These instruments use successive-approximation converters to perform the digitilization (analog-to-digital conversion). A simplified block diagram of such a DVM is shown in Figure

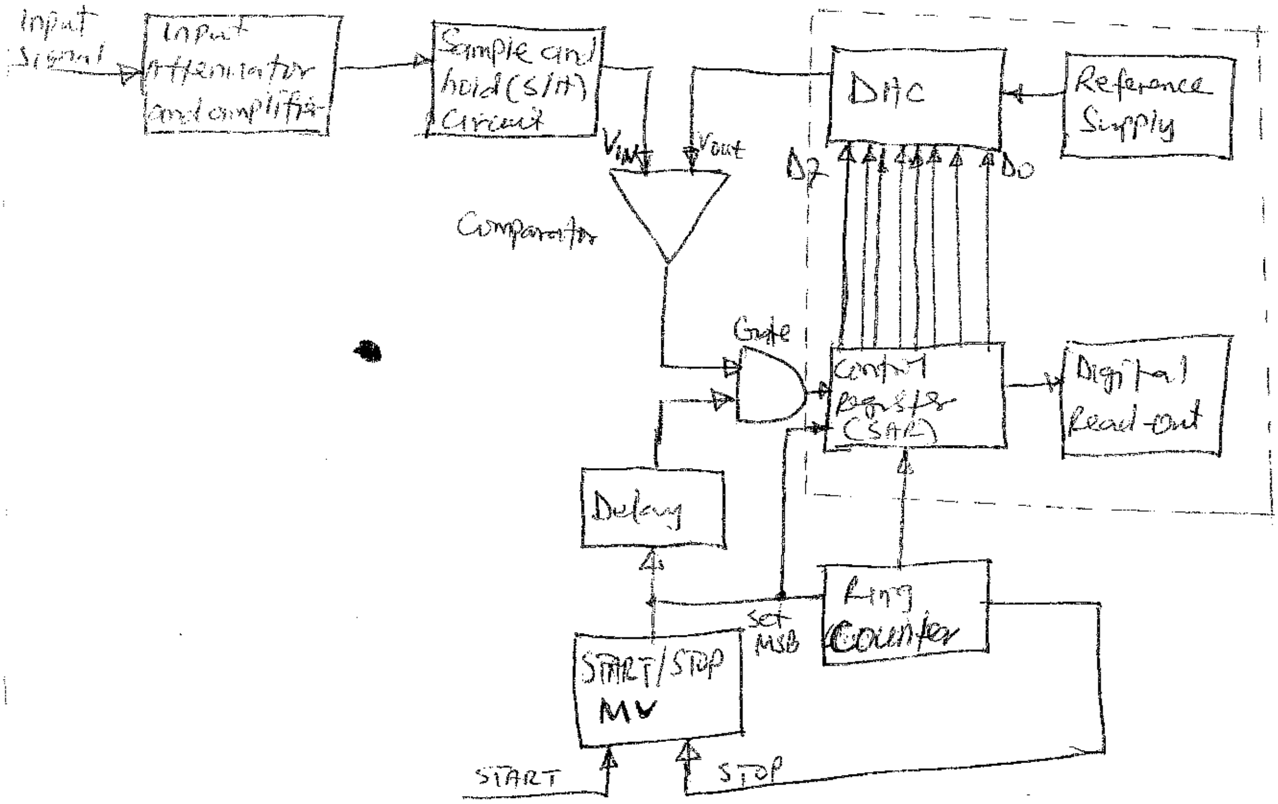


Figure Simplified block diagram of successive approximation DVM

At the start of the measurement cycle, a start pulse is applied to the start/stop multivibrator (MV). This sets a 1 in the most significant digit (MSB) of the control register and 0 in all the bits of less significance. Assuming an 8-bit control register, its reading would then be 10000000. This initial setting of the control register causes the output of DAC to be one-half the reference voltage i.e $\frac{1}{2}V_{REF}$. This converter output is compared with the unknown input by the comparator. In case the input voltage exceeds the converter reference voltage, the comparator produces an output that causes the control register to retain the 1 setting in MSB, and the converter continues to supply its reference output voltage of $\frac{1}{2} V_{REF}$.

The ring counter then advances one count, shifting a 1 in the second MSB of the control register and its reading becomes 11000000. This causes DAC to increase its reference output by 1 increment to $\frac{1}{4} V_{REF}$ i.e. $\frac{1}{2} V_{REF} + \frac{1}{4} V_{REF}$, and another comparison with the unknown input take place. If in this case the total reference voltage exceeds the unknown voltage, the comparator produces an output that makes the control register to reset its second MSB to 0. The converter output then returns to its previous value and awaits another input from the control register (SAR) for the next approximation. When the ring counter advances by 1, the third MSB is set to 1 and the converter output rises by the next increment of $\frac{1}{2} V_{REF} + \frac{1}{8} V_{REF}$. The measurement cycle thus proceeds through a series of successive approximations, retaining or rejecting the converter output in the manner explained. Finally, when the ring counter reaches its last count, the measuring cycle stops, and the digital output of the control register represents the final approximation of the unknown voltage.

With input voltages other than d.c, the input level changes during digitilization and decisions made during conversion are not consistent. However, this can be avoided by using a sample and holding (S/H) circuit and placing it in the input, directly following the input attenuator and amplifier.

WAVE ANALYZERS

A wave analyzer is an instrument designed to measure the relative amplitudes of single-frequency components in a complex or distorted waveform. The instruments acts as a frequency-selective voltmeter which is tuned to the frequency of one signal component while rejecting all the other signal components.

There are three basic instruments for analyzing harmonics of any periodic non-sinusoidal signal. These instruments are harmonic distortion analyzers; wave analyzers and spectrum analyzers.

Block diagram of wave analyzer is shown in figure below.

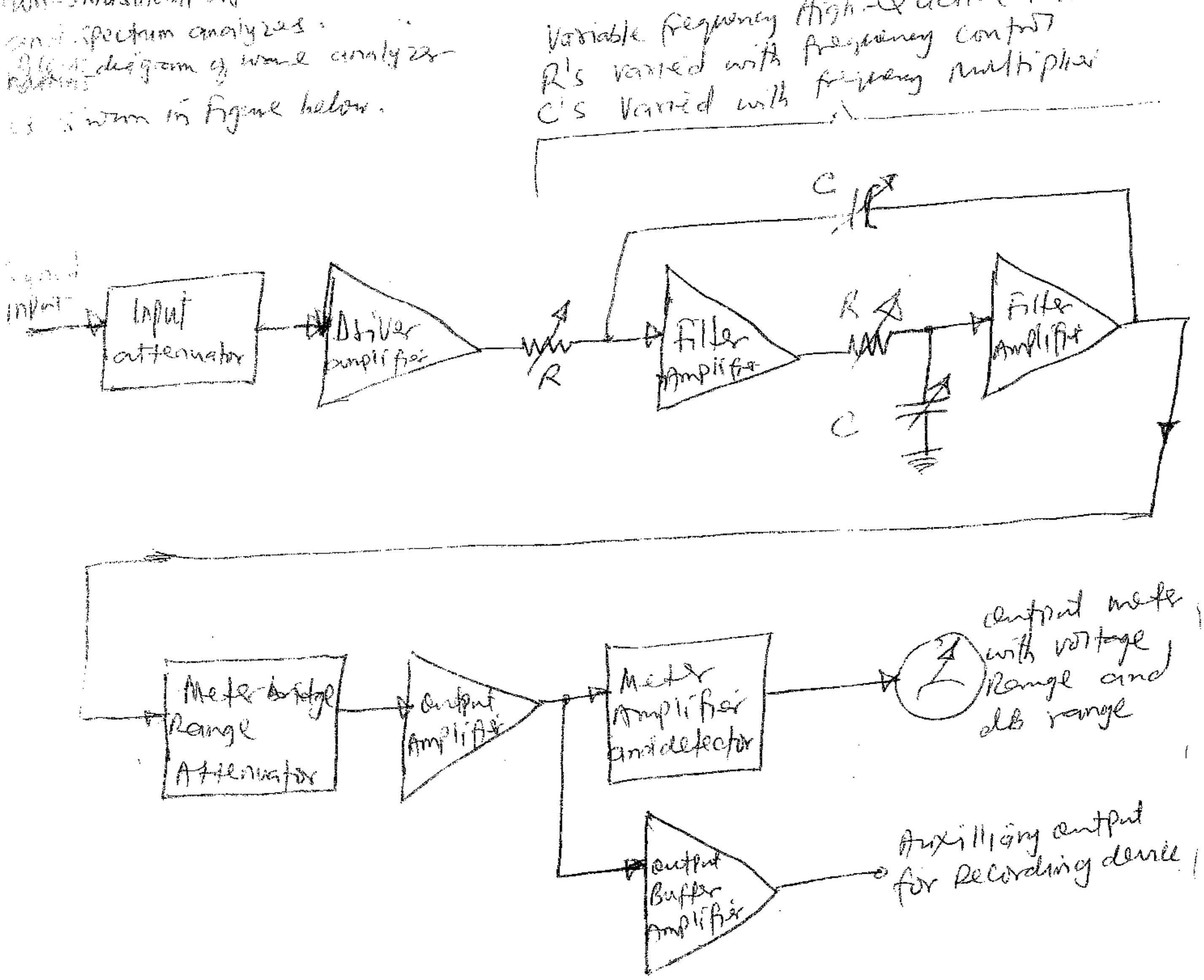


Figure Block diagram of Wave analyzer

The waveform to be analyzed in terms of its separate frequency components is applied to an attenuator that is set by the meter range switch on the front panel. A driver amplifier feeds the attenuated waveform to a high-Q-active filter. The filter consists of a cascaded arrangement of RC resonant sections and filter-amplifiers. Precision potentiometers are used to tune the filter to any desired frequency within the selected passband.

A final amplifier stage supplies the selected signal to the meter circuit and to an untuned buffer amplifier. The buffer amplifier can be used to drive a recorder or an electronic counter. The meter is driven by an average-type detector and usually has several voltage ranges as well as a decibel scale.

Applications of the wave analyzer are found in the fields of electrical measurements, ... sound and vibration analysis. The wave analyzer is applied industrially in the field of reduction of sound and vibration generated by machines and appliances

HARMONIC DISTORTION ANALYZERS

HARMONIC DISTORTION - In the ideal case, application of a sinusoidal input signal to an electronic device, such as an amplifier, should result in the generation of a sinusoidal output waveform. However, the output waveform is not exact replica of the input waveform because various types of distortion may arise. Distortion may be a result of the inherent non-linear characteristics of the transistors in the circuit or of the circuit components themselves. Nonlinear behaviour of circuit elements introduces harmonics of the fundamental frequency in the input waveform, and the resultant distortion is often referred to as harmonic distortion (HD).

Several methods have been devised to measure the harmonic distortion caused either by a single harmonic or by the sum of all the harmonics. Some of these methods are:

(i) Tuned-circuit harmonic analyzer

(ii) Heterodyne harmonic analyzer or wavemeter

(iii) Fundamental-suppression harmonic distortion analyzer.

SPECTRUM ANALYZER

The most common way of observing signals is to display them on an oscilloscope, with time as the X-axis (i.e amplitude of the signal versus time). This is the time domain. It is also used to display signals in the frequency domain. The instrument providing this frequency domain view is called the spectrum analyzer.

In a spectrum analyzer, the signals are broken down into their individual frequency components and displayed along X-axis of the CRO which is calibrated in terms of frequency. Thus the signal amplitude is displayed versus frequency on the CRO screen.

The block diagram of a basic spectrum analyzer is shown in Figure.

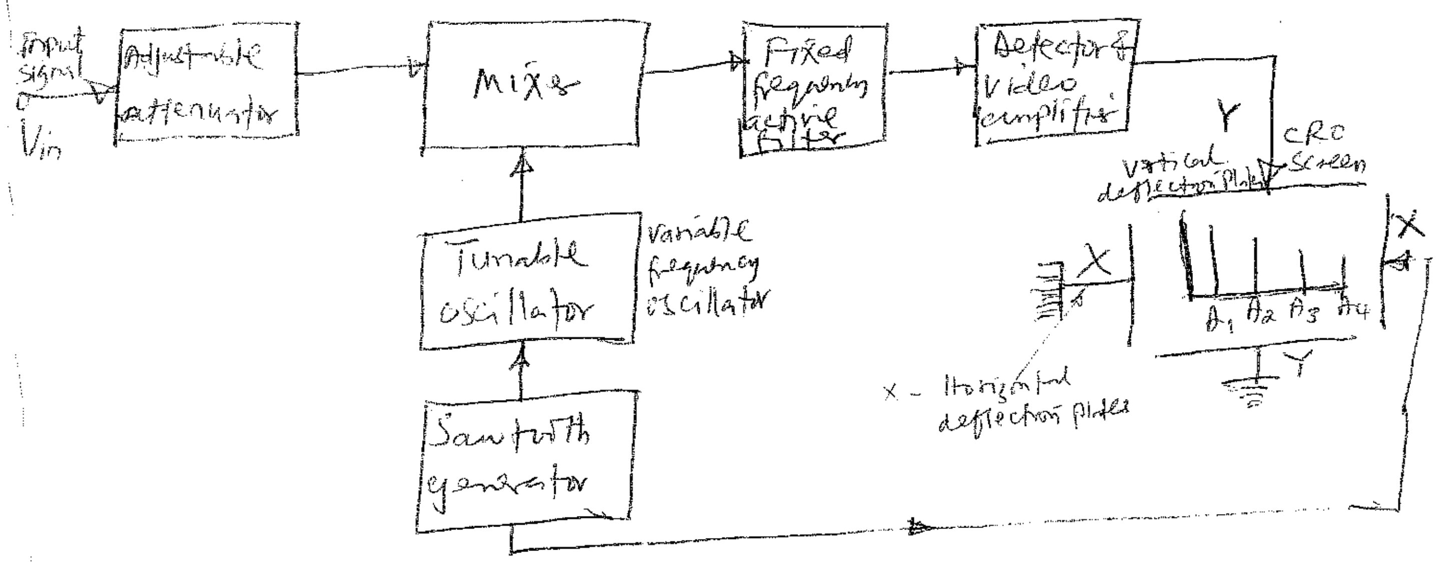


Figure Block diagram of a basic spectrum analyzer input signal is passed through an adjustable attenuator and then mixed it in a mixer with the signal from a variable frequency oscillator. The mixed signal is then filtered in the fixed frequency active filter. The output of the filter is then detected and supplied to the vertical deflection plates of a CRO. Sawtooth signal generator supplies the signal of the oscillator and the frequency of the oscillator is controlled by the instantaneous value of the sawtooth voltage.

The oscillator frequency sweeps linearly and increases from its minimum value to its maximum value as the sawtooth voltage rises from its minimum value to its maximum value. The same sawtooth signal is supplied to the horizontal deflection plates of the CRO, so it can be said that when sawtooth signal starts to rise from its minimum ~~signal~~ value, two events occur at the same time. First, frequency of oscillations starts to increase and secondly, spot on the screen starts travelling in horizontal direction.

As the oscillator frequency increases, firstly the fundamental components of the input signal is filtered out from the fixed frequency active filter. This filtered signal is then detected, amplified upto proper level and supplied to the vertical deflection plates of the CRO. So the spot on the screen is deflected to the left, point A_1 as shown in the figure, because sawtooth signal is at the minimum level, and upward, because output of detector and amplifier is supplied to the vertical deflection plates.

Output voltage of detector and amplifier is proportional to r.m.s value of filtered signal which itself is proportional to the r.m.s value of harmonic components. So the deflection of the spot on the screen in vertical direction i.e. shows the amplitude or rms value of that particular component.

As voltage of sawtooth signal rises further frequency of oscillator increases so second harmonic components is filtered out and after detection and amplification reaches the vertical deflection plates of CRO. Meanwhile increase in voltage of sawtooth signal causes little movement of the spot towards right side from its previous position in horizontal direction on the screen, so the spot comes to point A_2 as shown in Figure. Again on point A_2, deflection of the spot in vertical direction would be in proportion to the rms value of second harmonic component. Similarly all of the harmonic components appear on the CRO screen.

MEASUREMENT OF NON-ELECTRICAL QUANTITIES

Transducers are employed for converting non-electrical quantities into electrical signal and this electrical signals is processed by some electrical or electronic circuit and finally supplied to some output device for indication and recording purposes.

MEASUREMENT OF LIQUID LEVEL

The measurement of liquid level is widely carried out for monitoring as well as measuring quantitatively the liquid content in tanks, vessels, reservoirs, or the height of the liquid column in open-channel streams and various other similar cases in industrial processes.

They are different ~~types~~ methods of liquid level measurement as follows:

1. Resistive method
(ii) Float method
(iii) Force-balanced diaphragm system
(iv) Bubbler or purge system
(v) Capacitive methods
(vi) Ultrasonic method
(vii) Nuclear method

1. Resistive method

This method is also known as the contact point type. The arrangement is shown in figure below. In this method mercury column is operated by the liquid column. A number of resistances of suitable values are placed at various levels of liquid. As the level of liquid rises, the mercury level also rises and shorts the successive resistances and so the resistances R decreases or current through indicator increases.

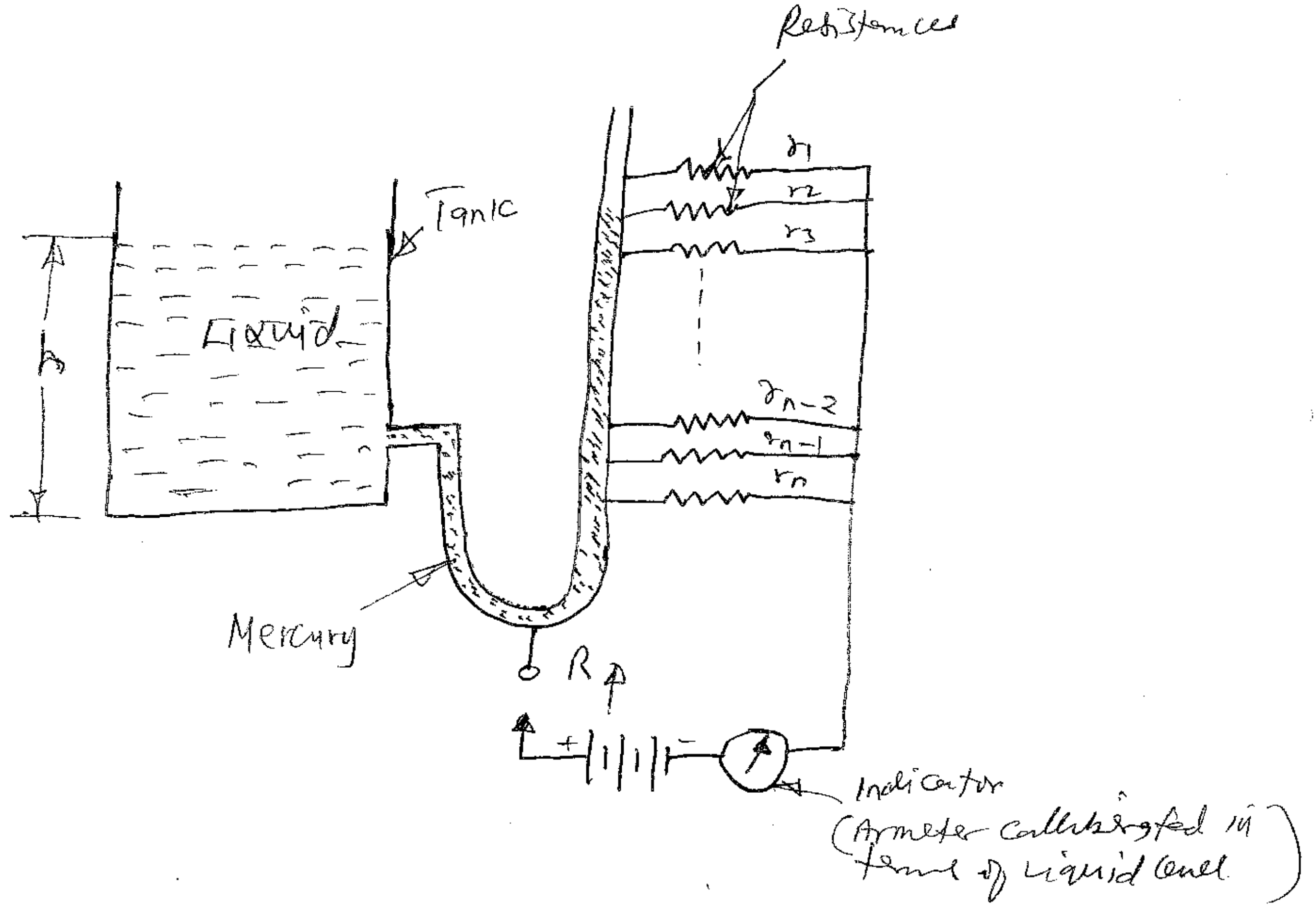


Figure Resistive or contact point type of measuring liquid level.

Resistances $r_1, r_2 \ldots r_n$ are so chosen that $1/R$ is a linear function of the liquid level.

2. CAPACITIVE METHODS

~~Liquid level~~, interface level, and the level of granular solids can be measured using the electrical capacitance effect. The capacitance device consists of a concentric type electrical ~~probe~~ inserted into the tank, and the liquid acts as the dielectric of the capacitor formed. The capacitance of the probe varies as the liquid level varies and electrical measurement of this capacitance gives the direct indication of the liquid level. The capacitance can be measured with an ac bridge system or as a change in frequency when coupled with a tank circuit of a high-frequency oscillator.

The arrangement for measurement of liquid level of a non-conducting fluid using capacitance effect is illustrated in Figure.

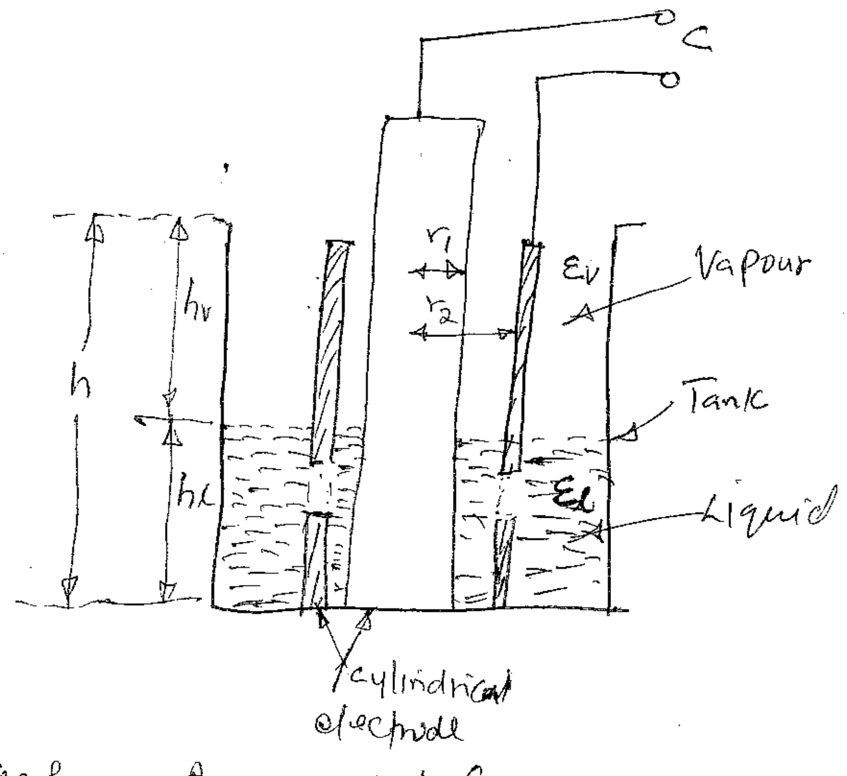


Figure Arrangement for measurement of level of a non-conducting fluid (Capacitive method)

The electrodes are arranged in the form of a pair of concentric cylinders in the tank with the holes in the outer cylinder at the lower end to allow passage of liquid.

As the level of the liquid rises, the capacitance of the capacitor between the electrode pair increases because the dielectric constant of vapour (ε_0 is very small in comparison to that of liquid (ε_l). The capacitance of the arrangement shown in the figure is given by the expression.

$$C = 2\pi\varepsilon_0 \frac{\varepsilon_l h_l + \varepsilon_v h_v}{\log_e r_2/r_1} \quad \ldots$$

Where ε_0 is the permittivity of free space in F/m

ε_l and ε_v are the relative permittivities of Liquid and Vapours respectively,

h_l and h_v are the heights of Liquid and Vapour Columns respectively in meters,

r_2 is the inner radius of outer cylinder &

r_1 is the outer radius of inner cylinder in m.

The outer cylinder is usually earthed in order to avoid stray capacitance effects.

MEASUREMENT OF THICKNESS

There are various methods of measuring thickness of sheets such as inductive, capacitive, ultrasonic and nuclear radiation.

Capacitive method – This method of measuring thickness is used in case the material to be tested is an insulator. The arrangement is shown in figure . Two metal plates are placed on either side of the sheet forming a parallel plate condenser. If the dielectric ~~constant~~ constant of the test sheet material is known, the spacing between the metal plates i.e the thickness of the sheet can be computed from the measured value of the capacitance of the arrangement.

DISPLACEMENT MEASUREMENT

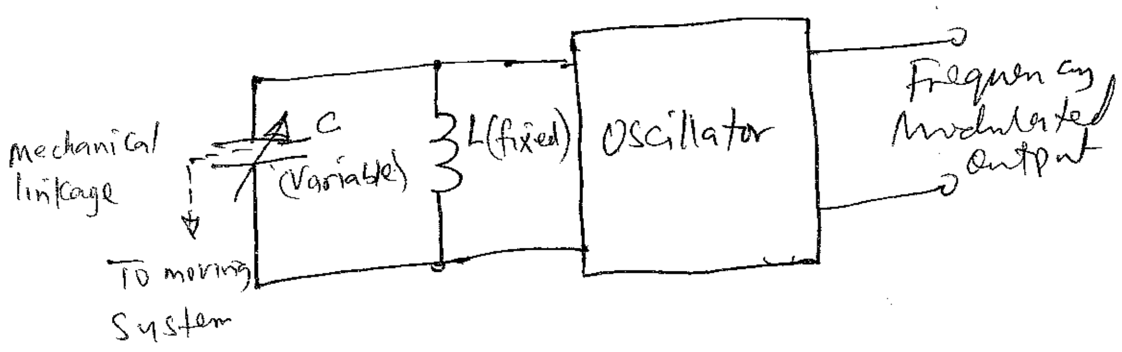
Introduction

Instrumental techniques are available for the measurement of linear as well as rotational displacements. The magnitude of measurement ranges from a few microns to a few centimetres in the case of linear displacement and a few seconds to 360° in the case of angular displacement.

Linear or translational displacement can be measured by a number of transducers such as resistive transducers, photoelectric transducers, LVDTs, capacitive transducers, piezoelectric transducers, photoelectric transducers, Hall effect transducers, or digital transducers.

Similarly rotational displacement can be measured by resistive potentiometers, inductive transducers, RVDTs, reluctance transducers, synchros, capacitive transducers or digital transducers.

Basic scheme for measurement of displacement – A scheme for measurement of displacement employing a capacitive transducer is illustrated in Figure



Figure

The basic scheme employs the basic elements of an LC oscillator whose frequency is affected by the variation in the capacitance of the capacitor. Movable plate of the capacitor is coupled to the moving system. So any movement of the moving system causes variation in the capacitor capacitance because of variation in separation between plates which in turn causes changes in the frequency of the oscillator. The change in frequency is a measure of the magnitude of the displacement under measurement.